# Advanced Time Series PSET 2

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## 1 Problem 1

Let

$$y_t = \rho y_{t-1} + u_t, \qquad u_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = \nu + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2, \qquad \varepsilon_t \overset{i.i.d.}{\sim} N(0,1).$$

Assume $\{y_t, \sigma_t\}$ is strictly stationary and $\sigma_t > 0$. Define the past information set

$$\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots).$$

Since

$$u_{t-j} = y_{t-j} - \rho y_{t-j-1} \qquad (j \geq 1),$$

the past values $\{u_{t-1}, u_{t-2}, \dots\}$ are measurable with respect to $\mathcal{F}_{t-1}$. Iterating the GARCH recursion backward therefore shows that $\sigma_t^2$ is also $\mathcal{F}_{t-1}$-measurable.

**(i)** We first have,

$$E(u_t^2) = E(\sigma_t^2 \varepsilon_t^2) = E(\sigma_t^2) E(\varepsilon_t^2) = E(\sigma_t^2),$$

since $\varepsilon_t$ is independent of $\sigma_t$ and $E(\varepsilon_t^2) = 1$.

Taking expectations in the GARCH recursion,

$$E(\sigma_t^2) = \nu + \alpha E(u_{t-1}^2) + \beta E(\sigma_{t-1}^2).$$

By stationarity,

$$E(\sigma_t^2) = E(\sigma_{t-1}^2) \equiv m, \qquad E(u_{t-1}^2) = E(\sigma_{t-1}^2) = m.$$

Thus,

$$m = \nu + \alpha m + \beta m,$$

so

$$E(\sigma_t^2) = \frac{\nu}{1 - \alpha - \beta},$$

provided $\alpha + \beta < 1$.

Next, compute the unconditional variance of $y_t$. Since

$$E(u_t \mid \mathcal{F}_{t-1}) = E(\sigma_t \varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t E(\varepsilon_t) = 0,$$

we have

$$E(y_t) = \rho E(y_{t-1}) + E(u_t).$$

By stationarity, $E(y_t) = E(y_{t-1})$, so

$$E(y_t) = 0.$$

Therefore,

$$\begin{aligned} Var(y_t) &= E(y_t^2) = E\big((\rho y_{t-1} + u_t)^2\big) \\ &= \rho^2 E(y_{t-1}^2) + 2\rho E(y_{t-1}u_t) + E(u_t^2). \end{aligned}$$

Now

$$E(y_{t-1}u_t) = E[y_{t-1}E(u_t \mid \mathcal{F}_{t-1})] = 0,$$

so

$$Var(y_t) = \rho^2 Var(y_{t-1}) + E(u_t^2).$$

Using stationarity again,

$$Var(y_t) = Var(y_{t-1}) \equiv \gamma_0,$$

thus,

$$\gamma_0 = \rho^2\gamma_0 + \frac{\nu}{1-\alpha-\beta}.$$

Therefore,

$$Var(y_t) = \frac{\nu}{(1-\rho^2)(1-\alpha-\beta)},$$

provided $|\rho| < 1$ and $\alpha + \beta < 1$.

**(ii)** Since

$$y_t = \rho y_{t-1} + u_t,$$

we have

$$E(y_t \mid \mathcal{F}_{t-1}) = \rho y_{t-1} + E(u_t \mid \mathcal{F}_{t-1}) = \rho y_{t-1}.$$

Thus,

$$E(y_t \mid y_{t-1}, y_{t-2}, \dots) = \rho y_{t-1}.$$

Also,

$$Var(y_t \mid \mathcal{F}_{t-1}) = Var(u_t \mid \mathcal{F}_{t-1}).$$

Because $\sigma_t$ is $\mathcal{F}_{t-1}$-measurable and $\varepsilon_t \sim N(0,1)$,

$$Var(u_t \mid \mathcal{F}_{t-1}) = Var(\sigma_t\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 Var(\varepsilon_t) = \sigma_t^2.$$

We then get

$$Var(y_t \mid y_{t-1}, y_{t-2}, \dots) = \sigma_t^2.$$

Since $u_{t-1} = y_{t-1} - \rho y_{t-2}$, we may also write

$$\sigma_t^2 = \nu + \alpha(y_{t-1} - \rho y_{t-2})^2 + \beta\sigma_{t-1}^2.$$

**(iii)** Conditional on $\mathcal{F}_{t-1}$,

$$u_t = \sigma_t \varepsilon_t \sim N(0, \sigma_t^2),$$

so

$$y_t \mid \mathcal{F}_{t-1} \sim N(\rho y_{t-1}, \sigma_t^2).$$

Therefore the conditional density of $y_t$ is

$$f(y_t \mid \mathcal{F}_{t-1}) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{(y_t - \rho y_{t-1})^2}{2\sigma_t^2}\right).$$

Thus, the conditional likelihood is

$$L(\rho, \nu, \alpha, \beta \mid y_0, y_{-1}, \sigma_0^2) = \prod_{t=1}^{T} \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{(y_t - \rho y_{t-1})^2}{2\sigma_t^2}\right),$$

where the conditional variances satisfy the recursion

$$\sigma_t^2 = \nu + \alpha(y_{t-1} - \rho y_{t-2})^2 + \beta\sigma_{t-1}^2, \qquad t = 1, \dots, T,$$

with initial condition

$$\sigma_0^2 = E(\sigma_t^2) = \frac{\nu}{1 - \alpha - \beta}.$$

In particular,

$$\sigma_1^2 = \nu + \alpha(y_0 - \rho y_{-1})^2 + \beta\sigma_0^2,$$

so the likelihood is indeed conditional only on $(y_0, y_{-1})$ and $\sigma_0^2$, exactly as requested in the problem.

Equivalently, the log-likelihood is

$$\ell(\rho, \nu, \alpha, \beta) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\left[\log \sigma_t^2 + \frac{(y_t - \rho y_{t-1})^2}{\sigma_t^2}\right].$$

We thus conclude taht,

$$E(\sigma_t^2) = \frac{\nu}{1 - \alpha - \beta}, \qquad Var(y_t) = \frac{\nu}{(1 - \rho^2)(1 - \alpha - \beta)},$$

$$E(y_t \mid y_{t-1}, y_{t-2}, \dots) = \rho y_{t-1}, \qquad Var(y_t \mid y_{t-1}, y_{t-2}, \dots) = \sigma_t^2,$$

and

$$L(\rho, \nu, \alpha, \beta \mid y_0, y_{-1}, \sigma_0^2) = \prod_{t=1}^{T}(2\pi\sigma_t^2)^{-1/2} \exp\left(-\frac{(y_t - \rho y_{t-1})^2}{2\sigma_t^2}\right)$$

with

$$\sigma_t^2 = \nu + \alpha(y_{t-1} - \rho y_{t-2})^2 + \beta\sigma_{t-1}^2, \qquad \sigma_0^2 = \frac{\nu}{1 - \alpha - \beta}.$$

# 2 Problem 2

We use the following design for our simulation:

$$y_t = 0.6\, y_{t-1} + 0.2\, y_{t-2} + u_t, \qquad u_t \overset{i.i.d.}{\sim} N(0,1), \qquad T = 200, \qquad p \in \{1,2,3\},$$

with 10,000 Monte Carlo repetitions and lag order chosen by BIC or AIC.

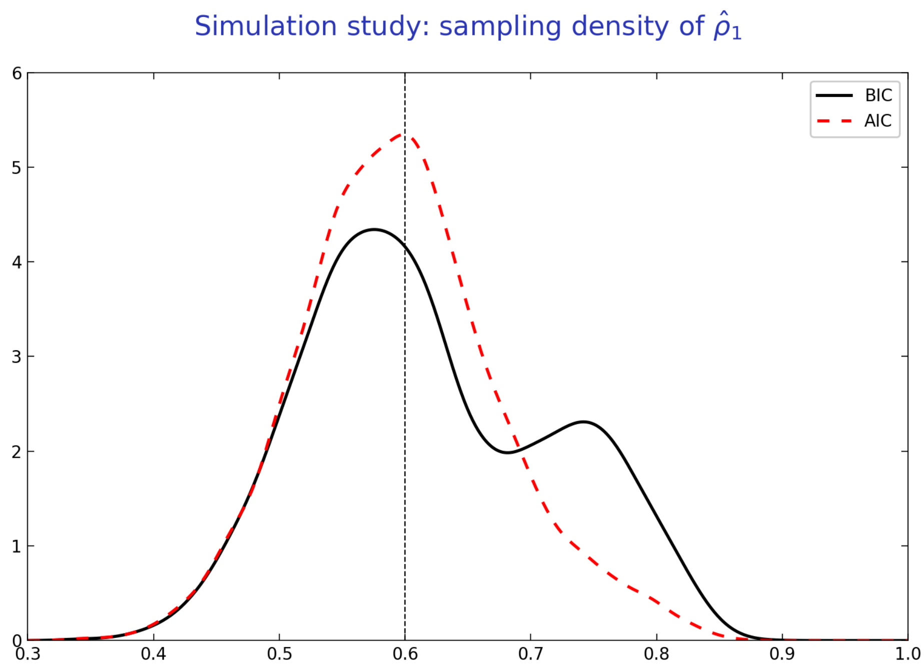


Figure 1: Replication of Slide 17: sampling density of the post-selection estimator $\hat{\rho}_1$.

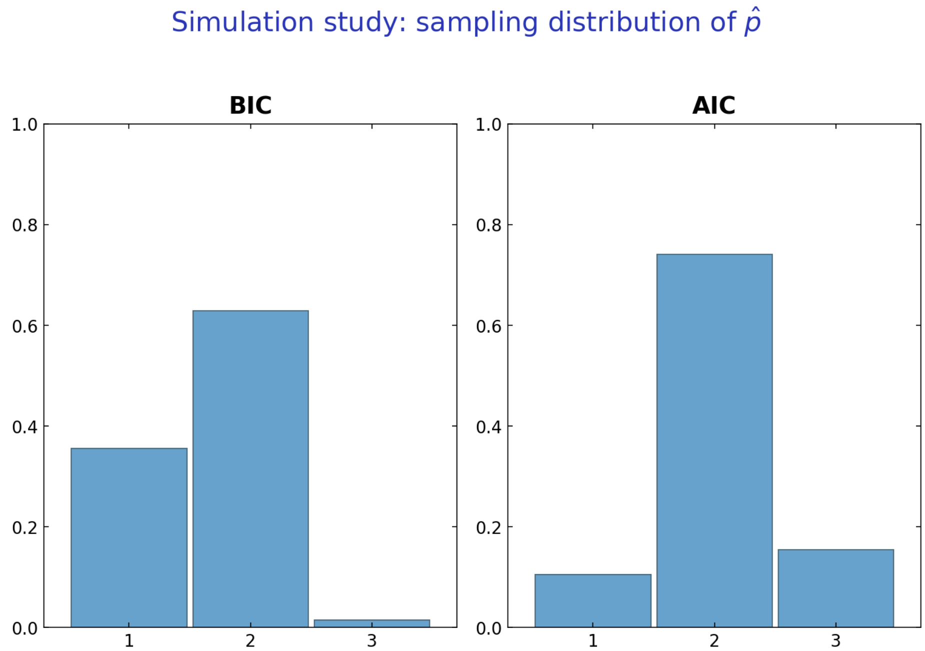


Figure 2: Replication of Slide 18: sampling distribution of the selected lag order $\hat{p}$.

The selection frequencies are

$$P(\hat{p}_{\text{BIC}} = 1, 2, 3) = (0.3562,\ 0.6291,\ 0.0147),$$

$$P(\hat{p}_{\text{AIC}} = 1, 2, 3) = (0.1046,\ 0.7411,\ 0.1543).$$

For the post-selection estimator of the first autoregressive coefficient,

$$E(\hat{\rho}_{1,\text{BIC}}) \approx 0.6195, \qquad \text{sd}(\hat{\rho}_{1,\text{BIC}}) \approx 0.0986,$$

$$E(\hat{\rho}_{1,\text{AIC}}) \approx 0.5960, \qquad \text{sd}(\hat{\rho}_{1,\text{AIC}}) \approx 0.0790.$$

We can thus see, as discussed in class, BIC is more conservative since it selects the too-small AR(1) model fairly often, which shifts the post-selection distribution of $\hat{\rho}_1$ to the right and makes it more dispersed. AIC selects the true AR(2) model more often, but it also over-selects AR(3) with nontrivial probability. In that sense, the Monte Carlo experiment shows very directly that once lag order is data-dependent, the distribution of the final estimator is a mixture over models rather than the distribution associated with a fixed AR(2) regression.

# 3 Problem 3

**(i)** Consider a univariate MA(1) process

$$y_t = u_t + \theta u_{t-1}, \qquad u_t \sim WN(0, \sigma_u^2).$$

Its autocovariances are

$$\Gamma_y(0) = \sigma_u^2(1 + \theta^2), \qquad \Gamma_y(1) = \sigma_u^2\theta, \qquad \Gamma_y(\ell) = 0 \text{ for all } |\ell| > 1.$$

Thus, its spectral density is

$$\begin{aligned} s_y(\omega) &= \frac{1}{2\pi}\sum_{\ell=-\infty}^{\infty}\Gamma_y(\ell)e^{-i\omega\ell} \\ &= \frac{1}{2\pi}\Big(\Gamma_y(0)+\Gamma_y(1)e^{-i\omega}+\Gamma_y(-1)e^{i\omega}\Big). \end{aligned}$$

Since the process is univariate and covariance stationary, we have

$$\Gamma_y(-1)=\Gamma_y(1),$$

so

$$s_y(\omega)=\frac{1}{2\pi}\Big(\Gamma_y(0)+2\Gamma_y(1)\cos\omega\Big).$$

Now a spectral density must satisfy

$$s_y(\omega)\geq 0 \qquad \text{for all } \omega\in[-\pi,\pi].$$

Therefore,

$$\Gamma_y(0)+2\Gamma_y(1)\cos\omega\geq 0 \qquad \text{for all } \omega\in[-\pi,\pi].$$

Since $\cos\omega\in[-1,1]$, the smallest possible value of $\Gamma_y(0)+2\Gamma_y(1)\cos\omega$ is

$$\Gamma_y(0)-2|\Gamma_y(1)|.$$

Thus nonnegativity of the spectral density implies

$$\Gamma_y(0)-2|\Gamma_y(1)|\geq 0,$$

or equivalently,

$$|\Gamma_y(1)|\leq\frac{\Gamma_y(0)}{2}.$$

Thus, there does not exist any univariate MA(1) process with

$$|\Gamma_y(1)|>\frac{\Gamma_y(0)}{2}.$$

As a check, using the MA(1) formulas above,

$$\frac{|\Gamma_y(1)|}{\Gamma_y(0)}=\frac{|\theta|}{1+\theta^2}\leq\frac{1}{2},$$

with equality if and only if $|\theta|=1$, which is consistent with the spectral argument.

**(ii)** There is no contradiction with the Wold decomposition theorem.

The Wold theorem says that any covariance stationary process can be written as the sum of a deterministic component and a purely nondeterministic component, where the latter has a moving-average representation of possibly infinite order:

$$y_t=d_t+\sum_{j=0}^{\infty}\psi_j\varepsilon_{t-j}, \qquad \sum_{j=0}^{\infty}\psi_j^2<\infty,$$

for a white noise process $\{\varepsilon_t\}$.

Thus, Wold does *not* say that every covariance stationary process is an MA(1). It only says that every covariance stationary process admits an MA($\infty$) representation (up to a deterministic component). Therefore, part (i) only shows that the class of univariate MA(1) processes is too restrictive to generate autocovariances with

$$|\Gamma_y(1)| > \frac{\Gamma_y(0)}{2}.$$

A covariance stationary process with such autocovariances may still exist; it simply cannot be represented as an MA(1). In general it would require a longer, typically infinite-order, moving-average representation.

Therefore the "paradox" is resolved by noting that

$$\text{Wold} \ \neq \ \text{MA}(1).$$

Wold guarantees an MA($\infty$) representation, not a finite MA(1) representation.

# 4 Problem 4

Let

$$y_t^{\dagger} \equiv E^*(y_t \mid \{x_\tau\}_{-\infty<\tau<\infty}) = \Theta(L)x_t, \qquad \Theta(z) \equiv \sum_{\ell=-\infty}^{\infty} \Theta_\ell z^\ell,$$

where $\Theta(z)$ is absolutely summable. Define the projection error

$$e_t \equiv y_t - y_t^{\dagger}.$$

Since $y_t^{\dagger}$ is the best linear predictor of $y_t$ from all leads and lags of $\{x_t\}$, the projection theorem implies that $e_t$ is orthogonal to the linear span of $\{x_\tau\}_{-\infty<\tau<\infty}$. In particular,

$$E(e_t x'_{t-h}) = 0 \qquad \text{for all } h \in \mathbb{Z}.$$

Equivalently,

$$\Gamma_{ex}(h) = 0 \qquad \text{for all } h \in \mathbb{Z}.$$

**(i)** Since

$$y_t = y_t^{\dagger} + e_t,$$

we have

$$\Gamma_{yx}(h) = \Gamma_{y^\dagger x}(h) + \Gamma_{ex}(h) = \Gamma_{y^\dagger x}(h) \qquad \text{for all } h \in \mathbb{Z}.$$

Now

$$y_t^{\dagger} = \sum_{\ell=-\infty}^{\infty} \Theta_\ell x_{t-\ell},$$

so

$$\begin{aligned}
\Gamma_{y^\dagger x}(h) &= Cov\left(\sum_{\ell=-\infty}^{\infty} \Theta_\ell x_{t-\ell},\, x_{t-h}\right) \\
&= \sum_{\ell=-\infty}^{\infty} \Theta_\ell Cov(x_{t-\ell}, x_{t-h}) \\
&= \sum_{\ell=-\infty}^{\infty} \Theta_\ell \Gamma_x(h-\ell).
\end{aligned}$$

Taking Fourier transforms,

$$\begin{aligned} s_{yx}(\omega) &= s_{y^\dagger x}(\omega) \\ &= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} e^{-i\omega h} \Gamma_{y^\dagger x}(h) \\ &= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} e^{-i\omega h} \sum_{\ell=-\infty}^{\infty} \Theta_\ell \Gamma_x(h-\ell). \end{aligned}$$

Using absolute summability to interchange the sums, and setting $m = h - \ell$, we get

$$\begin{aligned} s_{yx}(\omega) &= \sum_{\ell=-\infty}^{\infty} \Theta_\ell e^{-i\omega\ell} \left( \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} e^{-i\omega m} \Gamma_x(m) \right) \\ &= \Theta(e^{-i\omega})\, s_x(\omega). \end{aligned}$$

Since $s_x(\omega)$ is nonsingular for every $\omega \in [-\pi, \pi]$, it follows that

$$\Theta(e^{-i\omega}) = s_{yx}(\omega) s_x(\omega)^{-1} \qquad \text{for all } \omega \in [-\pi, \pi].$$

**(ii)** Since $y_t^\dagger = \Theta(L) x_t$, the spectrum of $y_t^\dagger$ is

$$s_{y^\dagger}(\omega) = \Theta(e^{-i\omega})\, s_x(\omega)\, \Theta(e^{-i\omega})^*.$$

Using the result from part (i),

$$\Theta(e^{-i\omega}) = s_{yx}(\omega) s_x(\omega)^{-1}.$$

Taking conjugate transposes and using that $s_x(\omega)$ is Hermitian,

$$\Theta(e^{-i\omega})^* = \left(s_x(\omega)^{-1}\right)^* s_{yx}(\omega)^* = s_x(\omega)^{-1} s_{xy}(\omega),$$

because $s_x(\omega)^* = s_x(\omega)$ implies $\left(s_x(\omega)^{-1}\right)^* = s_x(\omega)^{-1}$, and

$$s_{xy}(\omega) = s_{yx}(\omega)^*.$$

Thus,

$$\begin{aligned} s_{y^\dagger}(\omega) &= \Theta(e^{-i\omega})\, s_x(\omega)\, \Theta(e^{-i\omega})^* \\ &= s_{yx}(\omega) s_x(\omega)^{-1}\, s_x(\omega)\, s_x(\omega)^{-1} s_{xy}(\omega) \\ &= s_{yx}(\omega) s_x(\omega)^{-1} s_{xy}(\omega). \end{aligned}$$

Therefore,

$$s_{y^\dagger}(\omega) = s_{yx}(\omega) s_x(\omega)^{-1} s_{xy}(\omega) \qquad \text{for all } \omega \in [-\pi, \pi].$$

**(iii)** Now suppose $n_y = n_x = 1$, so all spectra are scalar-valued. The coherency is defined by

$$K_{yx}(\omega) \equiv \frac{s_{yx}(\omega)}{\sqrt{s_y(\omega) s_x(\omega)}}.$$

Since $s_x(\omega)$ is nonsingular and scalar, we have $s_x(\omega) > 0$, and by assumption $s_y(\omega) > 0$. From part (ii),

$$s_{y^\dagger}(\omega) = s_{yx}(\omega) s_x(\omega)^{-1} s_{xy}(\omega).$$

In the scalar case,

$$s_{xy}(\omega) = \overline{s_{yx}(\omega)},$$

so

$$s_{y^\dagger}(\omega) = \frac{|s_{yx}(\omega)|^2}{s_x(\omega)}.$$

Dividing by $s_y(\omega)$,

$$\frac{s_{y^\dagger}(\omega)}{s_y(\omega)} = \frac{|s_{yx}(\omega)|^2}{s_y(\omega)s_x(\omega)}.$$

But

$$|K_{yx}(\omega)|^2 = \left|\frac{s_{yx}(\omega)}{\sqrt{s_y(\omega)s_x(\omega)}}\right|^2 = \frac{|s_{yx}(\omega)|^2}{s_y(\omega)s_x(\omega)}.$$

Thus,

$$|K_{yx}(\omega)|^2 = \frac{s_{y^\dagger}(\omega)}{s_y(\omega)} \qquad \text{for all } \omega \in [-\pi, \pi].$$

In summary,

$$\Theta(e^{-i\omega}) = s_{yx}(\omega)s_x(\omega)^{-1},$$
$$s_{y^\dagger}(\omega) = s_{yx}(\omega)s_x(\omega)^{-1}s_{xy}(\omega),$$

and, in the scalar case,

$$|K_{yx}(\omega)|^2 = \frac{s_{y^\dagger}(\omega)}{s_y(\omega)}.$$

This shows that squared coherency is the frequency-by-frequency fraction of the spectrum of $y_t$ explained by the best linear predictor based on all leads and lags of $x_t$.

# 5 Problem 5

I used the monthly FRED unemployment-rate series `UNRATE` and `UNRATENSA`, restricted to 1948:1–2019:12. I compare a parametric AR spectral estimator with a smoothed-periodogram estimator to assess how much seasonal power remains after adjustment, especially near the annual frequency $\omega = \pi/6$.

**(i)** For each series I estimate an AR($p$) with intercept by least squares and choose $p \in \{1, \ldots, 50\}$ by BIC. The implied spectral estimator is

$$\widehat{s}_y^{(p)}(\omega) = \frac{\widehat{\sigma}^2}{2\pi\left|1 - \sum_{\ell=1}^{p} \widehat{A}_\ell e^{-i\omega\ell}\right|^2}, \qquad \omega \in [0, \pi].$$

I report the estimated log spectrum together with pointwise confidence bands based on delta-method standard errors, using the usual large-sample approximation that $\widehat{\sigma}^2$ is asymptotically independent of the estimated AR coefficients. BIC selects $p = 25$ for the unadjusted series and $p = 5$ for the adjusted series. The resulting AR log spectra are shown in Figure 3.

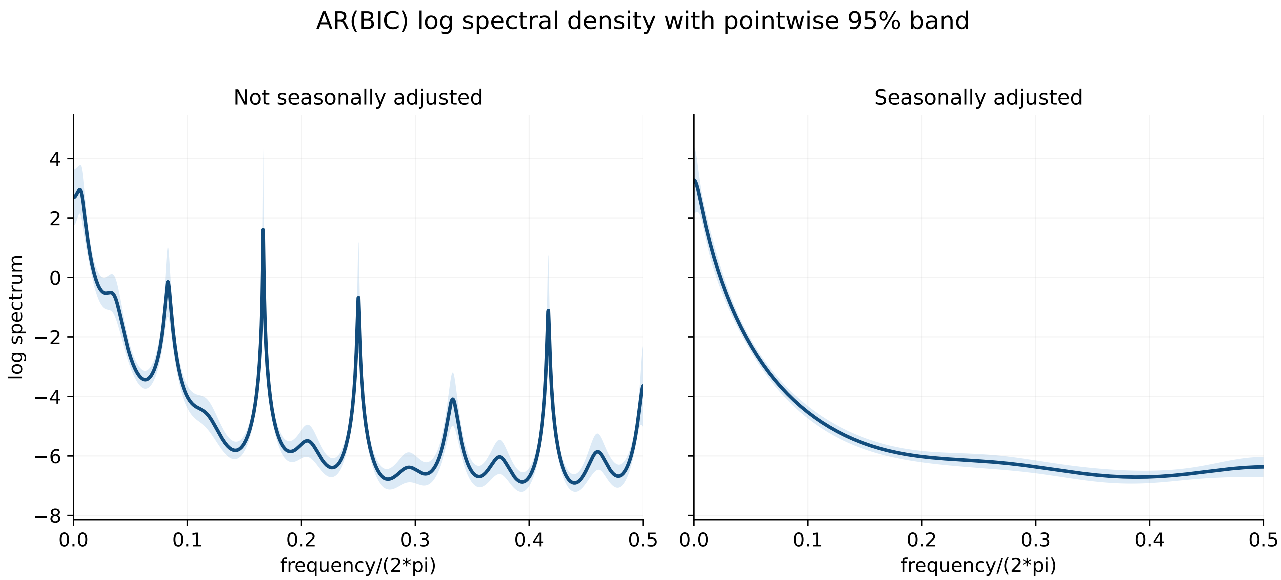


Figure 3: AR(BIC) log spectral density estimates with pointwise 95% confidence bands.

**(ii)** I also estimate the spectrum nonparametrically by smoothing the periodogram with the Epanechnikov kernel

$$w(x) = 1 - x^2, \qquad |x| \le 1,$$

using bandwidth $B = 10$ ordinates. This provides a less model-dependent check on the same spectral features. The estimator is

$$\widehat{s}_y(\omega_j) = \frac{1}{2\pi} \sum_{|k|\le B} w_B(k)\, \widehat{I}(\omega_{j+k}), \qquad w_B(k) = \frac{w(k/B)}{\sum_{|m|\le B} w(m/B)}.$$

For the log spectrum I use the asymptotic normal approximation from the notes,

$$\frac{\log \widehat{s}_y(\omega_j) - \log s_y(\omega_j)}{\sqrt{\sum_{|k|\le B} w_B(k)^2}} \overset{a}{\sim} N(0,1),$$

which yields a pointwise 95% confidence band. The smoothed log spectra are shown in Figure 4.

**(iii)** Both estimators deliver the same basic conclusion: the unadjusted series has a pronounced concentration of power around the annual seasonal frequency $\omega = \pi/6$, whereas the adjusted series is much flatter in that region. The comparison at $\omega = \pi/6$ is summarized below:

| Series | BIC lag | AR log spectrum at $\pi/6$ | Kernel log spectrum at $\pi/6$ |
|---|---|---|---|
| Not seasonally adjusted | 25 | -0.18 | -1.03 |
| Seasonally adjusted | 5 | -3.97 | -5.54 |

At $\omega = \pi/6$, the unadjusted series has 3.79 more log spectral power under the AR estimator and 4.51 more under the kernel estimator. In both figures this appears as a clear hump at the 12-month cycle for `UNRATENSA` that is largely absent for `UNRATE`. On that basis, the seasonal adjustment appears to remove the annual component effectively while leaving the lower-frequency business-cycle variation intact.

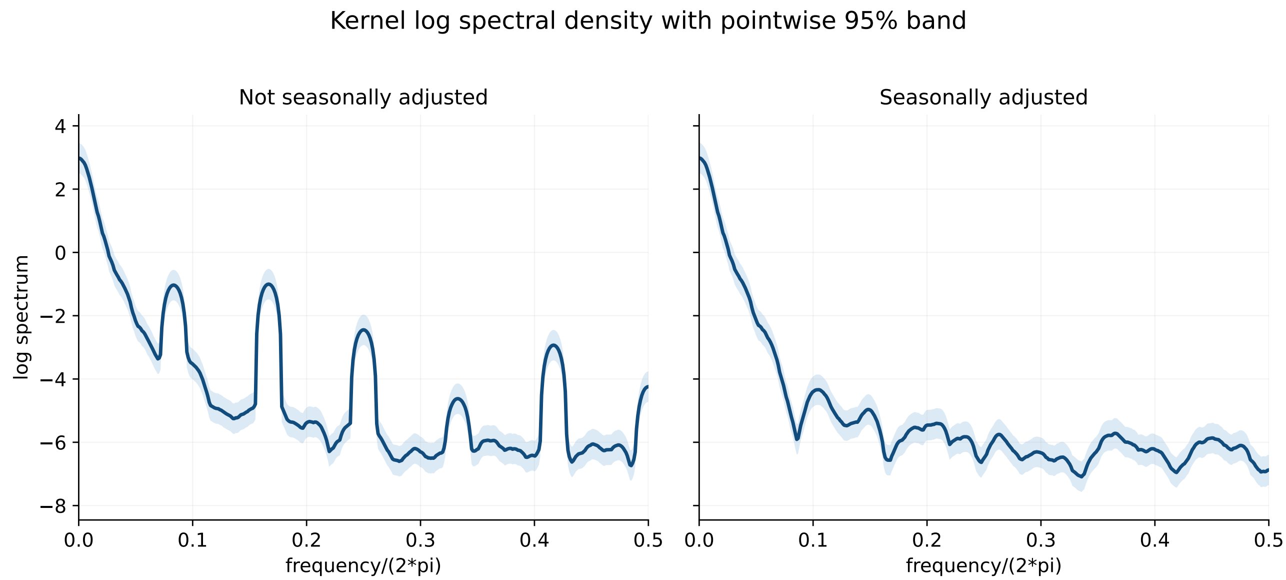


Figure 4: Kernel log spectral density estimates with pointwise 95% confidence bands.

# 6 Problem 6

For the NKPC estimation I use the seasonally adjusted FRED series `GDPDEF` and `UNRATE`. Inflation is measured by quarter-over-quarter log growth of the GDP deflator,

$$\pi_t = \Delta \log(\mathrm{GDPDEF}_t),$$

and the slack variable $x_t$ is the quarterly average unemployment rate. The largest overlapping quarterly sample before 2020 is 1948Q1–2019Q4 (288 quarters). After constructing the required leads and lags, the effective estimation sample becomes 1948Q4–2019Q3 (284 observations).

**(i)** Imposing rational expectations, I estimate the hybrid NKPC through the moment conditions

$$E\big[\big(\pi_t - c - \gamma_f \pi_{t+1} - \gamma_b \pi_{t-1} - \lambda x_t\big) Z_{t-1}\big] = 0,$$

where

$$Z_{t-1} = \big(1, \pi_{t-1}, \pi_{t-2}, \pi_{t-3}, x_{t-1}, x_{t-2}, x_{t-3}\big)'.$$

Writing

$$u_t(\theta) = \pi_t - c - \gamma_f \pi_{t+1} - \gamma_b \pi_{t-1} - \lambda x_t, \qquad g_t(\theta) = Z_{t-1} u_t(\theta),$$

the GMM estimator solves

$$\widehat{\theta}(W) = \arg\min_{\theta} \bar{g}(\theta)' W \bar{g}(\theta), \qquad \bar{g}(\theta) = \frac{1}{T} \sum_{t=1}^{T} g_t(\theta).$$

Using the weight matrix

$$\widehat{W}_1 = \left(\frac{Z'Z}{T}\right)^{-1}$$

gives the 2SLS estimator requested in the question. Because the moments are linear in the parameters, the estimator has the closed form

$$\widehat{\theta}(\widehat{W}) = \left(\frac{X'Z}{T} \widehat{W} \frac{Z'X}{T}\right)^{-1} \left(\frac{X'Z}{T} \widehat{W} \frac{Z'y}{T}\right).$$

**(ii)** I then compute the efficient two-step GMM estimator using

$$\widehat{W}_2 = \widehat{\Omega}(\widehat{\theta}_1)^{-1},$$

where $\widehat{\Omega}(\theta)$ is a Bartlett/Newey-West HAC estimator of the long-run variance of $g_t(\theta)$:

$$\widehat{\Omega}(\theta) = \widehat{\Gamma}_0 + \sum_{\ell=1}^{L_T} \left(1 - \frac{\ell}{L_T + 1}\right) (\widehat{\Gamma}_\ell + \widehat{\Gamma}'_\ell),$$

$$\widehat{\Gamma}_\ell = \frac{1}{T} \sum_{t=\ell+1}^{T} \tilde{g}_t(\theta)\tilde{g}_{t-\ell}(\theta)', \qquad \tilde{g}_t(\theta) = g_t(\theta) - \bar{g}(\theta).$$

I use the automatic lag-truncation rule

$$L_T = \left\lfloor 4(T/100)^{2/9} \right\rfloor = 5,$$

which is a standard weak-dependence consistent choice.

**(iii)** For both estimators I report HAC-robust standard errors from the usual GMM sandwich formula

$$\widehat{\mathrm{Var}}(\widehat{\theta}) = \frac{1}{T}(D'\widehat{W}D)^{-1}\big(D'\widehat{W}\widehat{\Omega}\widehat{W}D\big)(D'\widehat{W}D)^{-1}, \qquad D = -\frac{Z'X}{T}.$$

The resulting estimates are:

| Parameter | 2SLS / one-step GMM | s.e. | Efficient GMM | s.e. |
|---|---|---|---|---|
| $c$ | -0.000159 | 0.000378 | -0.000255 | 0.000350 |
| $\gamma_f$ | 0.8225 | 0.1657 | 0.8363 | 0.1308 |
| $\gamma_b$ | 0.2069 | 0.1257 | 0.1937 | 0.0994 |
| $\lambda$ | -0.000013 | 0.000063 | 0.000001 | 0.000059 |

The estimates are stable across the two weighting matrices. In both specifications the forward-looking coefficient is large, around 0.84, the backward-looking coefficient is around 0.19 to 0.21, and the slope coefficient on unemployment is extremely small and statistically insignificant.

**(iv)** Using the efficient two-step estimator, Hansen's over-identification statistic is

$$J = T\,\bar{g}(\widehat{\theta}_2)'\widehat{\Omega}(\widehat{\theta}_2)^{-1}\bar{g}(\widehat{\theta}_2).$$

Since there are $r = 7$ instruments and $k = 4$ parameters, the null distribution is asymptotically $\chi^2_{r-k} = \chi^2_3$. In this sample,

$$J = 1.6043, \qquad p\text{-value} = 0.6584.$$

Thus the over-identifying restrictions are not rejected. Taken together with the parameter estimates, the evidence here points to a Phillips curve that is primarily forward-looking, with some backward-looking persistence, but with very little slope when slack is measured by the unemployment rate.